

Spectral-Separable Self-Attention for Efficient EEG Motor Imagery Classification

SMT-S3A: Frequency-Aware Attention with Linear Complexity

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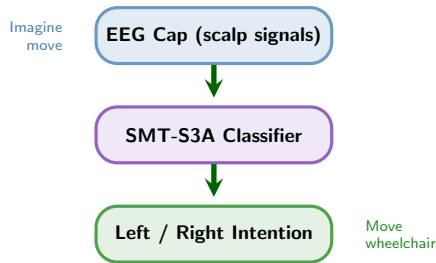
The Problem: Giving Paralyzed Patients Their Voice Back

Motor Imagery BCI:

- Patients with paralysis **imagine** moving their hand
- Electrodes on the scalp pick up brain signals (EEG)
- A classifier decodes *left vs. right* intent
- Output controls a wheelchair, prosthetic, or cursor

Why is deployment hard?

- Every person's brain signals are **different**
- A model trained on Person A *fails* on Person B
- Real systems must work **without per-user retraining**
- This is the *leave-one-subject-out* (LOSO) challenge



Goal: robust cross-subject decoding with minimal parameters.

Existing Approaches: What Works and What Doesn't

Generation 1 — CNNs:

- ShallowConvNet, EEGNet, FBCNet
- Fast, lightweight
- ✗ Treat channels independently — miss temporal context
- ✗ FBCNet uses *fixed* freq. bands (hard-coded)

Generation 2 — Transformers:

- ATCNet, SMT — model temporal dependencies
- Better accuracy on most benchmarks
- ✗ **Quadratic cost** $O(N^2D)$ — heavy on edge devices
- ✗ **Frequency-blind** — mix all EEG bands together

Table 1: LOSO accuracy on PhysioNet (105 subjects)

Model	Acc (%)	Freq-aware?
ShallowConvNet	71.20	No
EEGNet	72.85	No
FBCNet	71.90	Fixed only
ATCNet	73.45	No
SMT	72.13	No
SMT-S3A (ours)	81.75	Yes, learned

+9.62% over nearest transformer baseline.

Our Key Insight: Attention Should Think in Frequency Bands

The Problem with Standard Attention

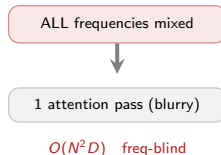
Standard transformer attention treats the EEG signal as a single undifferentiated stream. It has **no idea** that the mu band and beta band carry completely different motor information.

What neuroscience tells us:

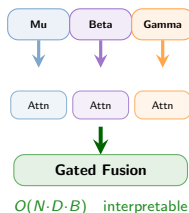
- **Mu (8–13 Hz):** drops when you imagine moving — *ERD*
- **Beta (13–30 Hz):** rises when motor cortex is idle — *ERS*
- **Gamma (30–38 Hz):** supplementary motor activity

Our idea: Decompose the signal into these bands first, *then* apply attention separately within each band.

Standard Attention



S3A (Ours)



What we build — SMT-S3A:

- ① **Patch Embedding:** Conv layers extract temporal features
- ② **S3A Block:** Spectral-Separable Self-Attention
 - Stage 1: Learnable Gaussian frequency decomposition
 - Stage 2: Separable attention per band
 - Stage 3: Gated cross-band fusion
- ③ **Log-Power Head:** Classification

Key claims we validate:

- Accuracy: **70.45% / 81.77% / 81.75%** on BCI2a / 2b / PhysioNet (LOSO)
- Efficiency: **98.6%** fewer attention FLOPs vs. SMT
- Interpretability: bands converge to **μ , β , γ** with *no supervision*
- Size: only **32K parameters**

SMT-S3A: Full Architecture

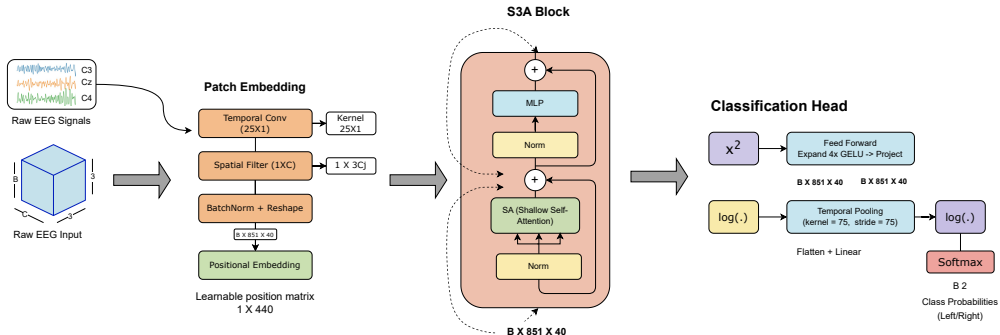
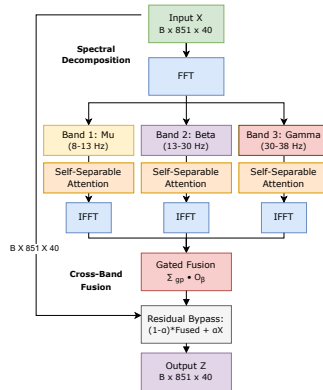


Figure 1: SMT-S3A architecture. Raw EEG $\rightarrow$ Convolutional patch embedding $\rightarrow$ S3A transformer block $\rightarrow$ Log-power classification head. Single transformer layer ($L=1$); dashed lines denote residual connections.

S3A Mechanism: Three Stages

S3A: Spectral-Separable Self-Attention Mechanism



Total Complexity: $O(N \cdot D \cdot B)$ where $B=3$

Figure 2: S3A forward pass. Stage 1: FFT $\rightarrow$ learnable Gaussian band decomposition. Stage 2: linear-complexity separable attention per band. Stage 3: softmax gating fusion with spectral residual bypass.

S3A Stage 1: Learnable Spectral Decomposition

Project tokens to frequency domain:

$$\tilde{\mathbf{X}} = \text{FFT}(\mathbf{X}, \text{dim}=1) \quad (1)$$

Learnable Gaussian band-pass masks ($b = 1, \dots, B$):

$$M_b(f) = \exp\left(-\frac{(f - \hat{\mu}_b)^2}{2\hat{\sigma}_b^2}\right), \quad f \in [0, 1] \quad (2)$$

Normalized soft partition (bands sum to 1):

$$\bar{M}_b(f) = M_b(f) / \sum_j M_j(f) \quad (3)$$

Band-limited reconstruction:

$$\mathbf{X}_b = \text{IFFT}(\tilde{\mathbf{X}} \odot \bar{M}_b) \quad (4)$$

Key Design Choice

Band centers $\hat{\mu}_b$ and widths $\hat{\sigma}_b$ are *differentiable parameters* — learned end-to-end by backprop. No manual frequency specification needed.

Band Diversity Regularizer:

$$\mathcal{L}_{\text{div}} = \frac{1}{|\mathcal{P}|} \sum_{(i,j) \in \mathcal{P}} \frac{1}{(\hat{\mu}_i - \hat{\mu}_j)^2 + \epsilon} \quad (5)$$

- Repels band centers apart
- Prevents all bands collapsing to same frequency
- Weight $\lambda_{\text{div}}=0.01$

S3A Stages 2 & 3: Attention and Fusion

Stage 2 — Separable Attention (per band b , $m=8$ heads):

Context scores [$O(N \cdot D)$ — no $N \times N$ matrix]:

$$\mathbf{c}_b = \text{softmax}_N(\mathbf{X}_b \mathbf{W}_I^{(b)}) \quad (1)$$

Compress tokens into a context vector:

$$\mathbf{v}_b^{(j)} = \sum_i c_b^{(j)}(i) \cdot (\mathbf{W}_K^{(b)} \mathbf{x}_{b,i})^{(j)} \quad (2)$$

Broadcast back to every token:

$$\mathbf{O}_b = \text{Concat}_j[\mathbf{v}_b^{(j)} \odot \text{ReLU}(\mathbf{W}_V^{(b)} \mathbf{x}_{b,i})^{(j)}] \quad (3)$$

Stage 3 — Gated Cross-Band Fusion:

$$\mathbf{Y}_{\text{fused}} = \sum_b \frac{e^{g_b}}{\sum_j e^{g_j}} \cdot \mathbf{O}_b \quad (4)$$

Spectral residual bypass:

$$\mathbf{Y} = \mathbf{W}_O[(1-\alpha)\mathbf{Y}_{\text{fused}} + \alpha\mathbf{X}] \quad (5)$$

$\alpha = \sigma(\tilde{\alpha}) \in [0, 1]$ — learned, prevents info loss

Total Complexity

$O(ND \log N + N \cdot D \cdot B)$

$B=3, \log N \approx 10 \Rightarrow$ **linear in N**

vs. SMT: $O(N^2 D)$ with $N=851$

Classification Head & Architecture Summary

Transformer Block (pre-norm, $L=1$):

$$\mathbf{Z}' = \text{S3A}(\text{LN}(\hat{\mathbf{F}})) + \hat{\mathbf{F}} \quad (1)$$

$$\mathbf{Z} = \text{FFN}(\text{LN}(\mathbf{Z}')) + \mathbf{Z}' \quad (2)$$

Log-Power Head:

$$\mathbf{Z}_{\log} = \log(\text{AvgPool}(\mathbf{Z}^2) + \epsilon) \quad (3)$$

$$\hat{\mathbf{y}} = \text{Softmax}(\mathbf{W}_c \text{Dropout}(\text{Flat}(\mathbf{Z}_{\log}))) \quad (4)$$

AvgPool: kernel 75, stride 15; Dropout 0.5

Table 2: Architecture at a glance

Component	Config
Electrodes	C3, Cz, C4 @ 250 Hz
Epoch	0.5–4.0 s (875 samples)
Temporal conv	$D=40$, kernel 25×1
Tokens	$N=851$
S3A bands	$B=3$
Heads per band	$m=8$
Transformer layers	$L=1$
Total params	32.1K

Classification Performance: LOSO Cross-Validation

Table 3: Comprehensive LOSO comparison. All methods use C3/Cz/C4 electrodes only.

Model	Dataset 2a (9 subj)		Dataset 2b (9 subj)		PhysioNet (105 subj)	Params
	Acc (%)	κ	Acc (%)	κ	Acc (%)	
ShallowConvNet	61.96	0.24	73.48	0.47	71.20	10.1K
EEGNet	63.50	0.27	74.34	0.49	72.85	2.6K
FBCNet	62.00	0.24	73.07	0.46	71.90	15.2K
ATCNet	61.57	0.23	74.61	0.49	73.45	37.1K
SMT	67.28	0.35	76.41	0.53	72.13	29.8K
SMT-S3A (ours)	70.45	0.409	81.77	0.635	81.75	32.1K
<i>vs. SMT</i>	+3.17	+0.059	+5.36	+0.105	+9.62	

Key: Largest gain on PhysioNet (+9.62%, 105 subjects) — strongest evidence of cross-subject generalization.

Per-Subject Results on BCI Competition IV

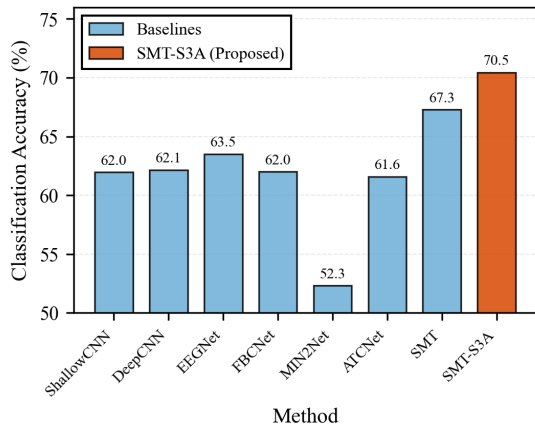


Figure 3: LOSO accuracy on BCI Competition IV Dataset 2a. SMT-S3A (70.45%) outperforms all baselines across most subjects.

Dataset 2a — Per-Subject (%):

Table 4: BCI Competition IV Dataset 2a fold-wise results

	S1	S2	S3	S4	S5	S6	S7	S8	S9	Avg
SMT	78.5	56.9	88.5	61.5	50.7	66.3	53.8	81.6	67.7	67.3
S3A	64.6	54.9	80.6	70.1	65.3	65.3	73.6	88.9	70.8	70.4

Dataset 2b — Per-Subject (%):

Table 5: BCI Competition IV Dataset 2b fold-wise results

	S1	S2	S3	S4	S5	S6	S7	S8	S9	Avg
SMT	78.1	81.3	88.9	72.5	68.8	71.3	82.5	70.0	74.3	76.4
S3A	85.0	83.8	93.8	78.1	75.6	77.5	87.5	76.9	78.1	81.8

Wins **6/9** folds on Dataset 2a, **7/9** on Dataset 2b.

Efficiency: 98.6% Reduction in Attention FLOPs

Table 6: Model efficiency comparison (batch size 1, GPU inference).

Model	Params	Attn FLOPs	Infer (ms)
ShallowConvNet	10.1K	–	0.54
EEGNet	2.6K	–	2.43
ATCNet	37.1K	–	15.68
SMT	29.8K	29.0M	5.08
SMT-S3A	32.1K	0.4M	5.12

98.6% FLOPs Reduction

29.0M $\rightarrow$ 0.4M attention FLOPs with *no inference latency penalty* (5.08 ms vs. 5.12 ms).

Why this matters for BCI:

- Fits on edge/embedded hardware
- Longer recording epochs become feasible
- Only +2.3K parameters over SMT
- 32K total — leaner than ATCNet (37.1K)

Ablation Study: Every Design Choice Contributes

Table 7: Ablation on BCI Competition IV Dataset 2a (LOSO).

Variant	Acc (%)	Δ vs. full
SMT (baseline)	67.28	-3.17
Separable only ($B=1$)	68.12	-2.33
SMT-S3A ($B=3$)	70.45	—
Too many bands ($B=5$)	69.21	-1.24
w/o diversity loss	67.89	-2.56
w/o residual bypass	68.75	-1.70
w/o mirror augment	68.02	-2.43
Mu band only	64.32	-6.13
Beta band only	62.85	-7.60
Gamma band only	58.91	-11.54
All 3 bands	70.45	—

What the ablation tells us:

- 1 **Spectral decomp gives +2.33%** — biggest single gain
- 2 $B=3$ maps naturally to mu/beta/gamma; $B=5$ overfits
- 3 Without diversity loss, all bands collapse to the same frequency
- 4 No single band alone is sufficient — they're *complementary*
- 5 Mirror augmentation helps by encoding left-right symmetry

The Interpretability Bonus: Model Rediscovered Neuroscience

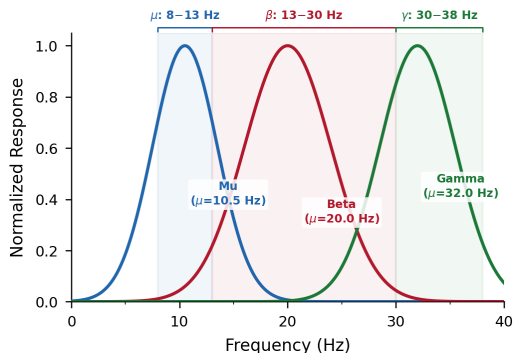


Figure 4: Learned band frequency responses after training on Dataset 2a (LOSO). Shaded regions: textbook EEG frequency bands. Bands converge to mu/beta/gamma **without any frequency-domain supervision**.

What the model learned autonomously:

- 1 **Band 1** → **Mu (10–12 Hz)**
ERD: power drops during motor planning
- 2 **Band 2** → **Beta (18–22 Hz)**
ERS: power rises during cortical rest
- 3 **Band 3** → **Low Gamma (30–38 Hz)**
Supplementary motor activity

Why This Matters

The model received only class labels. Yet it discovered the same frequency boundaries that took neuroscientists decades to establish — validating the architecture's design.

Conclusion: Three Things in One Architecture

Main Message

Domain knowledge (EEG has band structure) encoded as an architectural inductive bias simultaneously improves **accuracy**, **efficiency**, and **interpretability**.

Key Contributions:

- 1 **S3A mechanism:** Gaussian spectral decomposition + per-band separable attention + gated fusion. $O(N^2D) \rightarrow O(N \cdot D \cdot B)$
- 2 **Emergent interpretability:** Band centers spontaneously converge to mu/beta/gamma — no supervision needed.
- 3 **State-of-the-art at 32K params:** +3.17 / +5.36 / +9.62% over SMT on BCI2a / 2b / PhysioNet.

Practical Impact:

- Real-time, edge-deployable BCI
- No trade-off: accuracy + efficiency + interpretability together

Future Directions:

- Seizure detection
- Sleep staging
- Cognitive state monitoring
- Any signal with known spectral structure

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Thank You!

Questions?

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